How-To Library

Informatica

Deploying the Informatica® Data Engineering Streaming 10.4.0 AWS Marketplace Solution

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Abstract

This deployment reference provides step-by-step instructions for deploying Informatica® Data Engineering Streaming on the Amazon Web Services Cloud Marketplace. Automated reference deployments use AWS CloudFormation templates to launch, configure, and run the AWS compute, network, storage, and other services required to deploy a specific workload on AWS.

Supported Versions

- Data Engineering Streaming 10.4

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Overview

Customers of Amazon Web Services and Informatica can launch a Data Engineering Streaming deployment from the AWS marketplace to create an Informatica domain in the AWS cloud and explore Data Engineering Streaming functionality.

Intended Audience

This guide is for users who are responsible for deploying the marketplace solution of Data Engineering Streaming 10.4.0 on the AWS Cloud.

As a user with administrator privileges to deploy applications on AWS, you should be familiar with AWS resources such as Cloud formation, VPC, EC2 instances, S3 storage, RDS, Internet gateway, NAT Gateway, route tables, security groups and roles, and Elastic IP. See the AWS documentation.

Familiarity with the concepts of IP CIDR, public and private IP addresses is recommended.

You should also be familiar with Informatica Data Engineering solutions. To find Data Engineering Streaming, see the Informatica documentation portal.

Costs and Licenses

You are responsible for the cost of the AWS services used while running this deployment. There is no additional cost for using this marketplace deployment.

The AWS CloudFormation template for this deployment includes configuration parameters that you can customize. Some of these settings, such as instance type, will affect the cost of deployment. See the pricing pages for each AWS service you will be using for cost estimates.

This deployment requires a license for Informatica Data Engineering Streaming. To sign up for a demo license, contact your organization's Informatica sales contact or Informatica Global Customer Support.

The following table lists the instance types that you can choose based on sizing requirements:

Virtual Machine	Instance Type
Database	M5.xlarge / R5.xlarge
Informatica domain	M4.2xlarge / M5.2xlarge
Bastion server	M5.xlarge / M4.xlarge

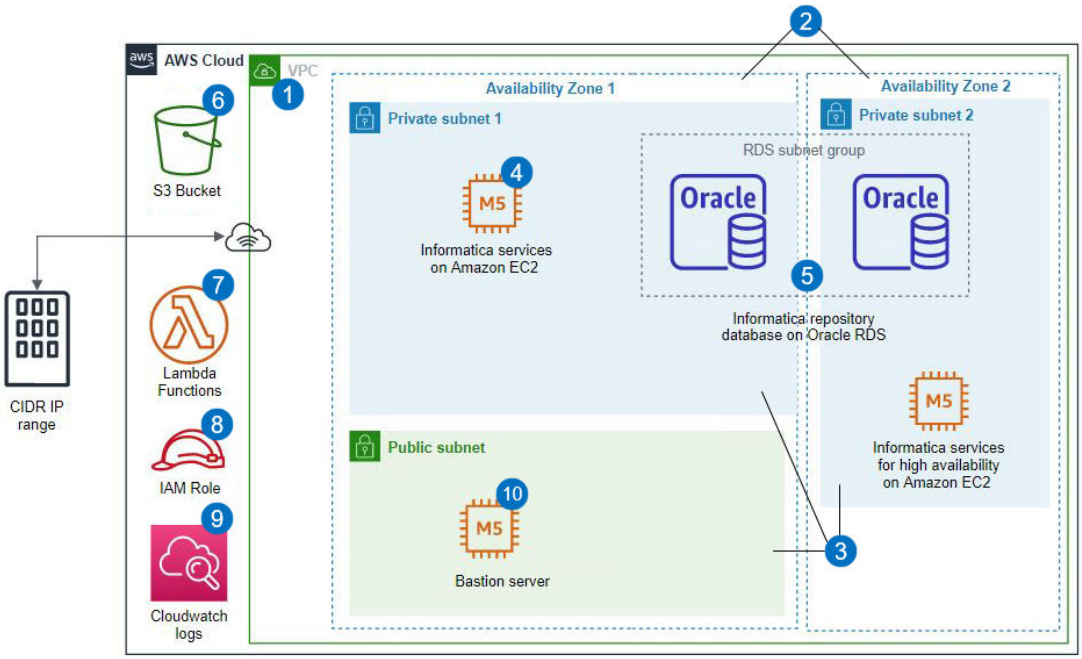
The deployment chooses M4 or M5 type of instances based on the instance type availability for the specific AWS region.

Architecture

When you deploy Data Engineering Streaming in a new VPC, AWS CloudFormation templates create and connect the following resources in the VPC:

- Informatica domain on an EC2 instance, with additional instances to contain nodes in the Data Integration Service grid
- Informatica clients on a remote Windows bastion server, on a public subnet
- Amazon S3 storage resources and connections for source and target data in existing S3 buckets
- Amazon RDS relational databases for Informatica domain repositories
- AWS security and account management services
- Lambda functions

The following image shows the architecture of the Data Engineering Streaming on AWS marketplace solution:



The numbers in the architecture diagram correspond to items in the following list:

1. A virtual public cloud (VPC) configured across two Availability Zones to contain the Data Engineering Streaming deployment.
2. Availability zones. For each Availability Zone, this deployment provisions one private and one public subnet.
3. Subnets to contain specific elements of the deployment. Creates two private subnets, plus one public subnet if you want to use a Windows bastion server for Informatica clients. Creates each of the subnets in a different availability zone.
4. The Informatica domain, including the Model Repository Service and the Data Integration Service.
5. Oracle database to contain the following Informatica repositories:
   - Domain configuration repository
   - Model repository, a relational database that stores all the metadata for projects created using Informatica client tools. The Model repository also stores run-time and configuration information for applications that are deployed to a Data Integration Service.
6. S3 storage, to act as a temporary location for files that the Data Integration Service moves between EC2 instances and the EMR cluster.
7. AWS Lambda functions.
8. IAM roles.
9. Amazon CloudWatch.
10. Informatica clients in a separate EC2 bastion server in a public subnet.

AWS Resources Created by the Deployment

The deployment process creates or includes the following components:

- Components created when you specify the solution to create a new VPC.
- Components created, as well as some that already exist, when you specify the solution to use an existing VPC.

- A bastion server, if you choose to create one.

Components in a Deployment on a New VPC

The following table lists components that the cloud formation template creates when you deploy the software in a new virtual private cloud (VPC):

Component	Number of Components Created
VPC	1
Internet gateway	1
Subnet	4
Route table	2
Security group	The following security groups: - One security group for the RDS - One security group for the Informatica domain instance - If you selected the option to create the bastion server, one security group to support remote login to the bastion server. - Based on your choice, two security groups for Informatica EMR workflow connections.
DB subnet group	1
RDS	1 Oracle RDS instance with 200 GB of storage capacity that runs on a db.m5.xlarge DB instance type.
EC2	Informatica domain with an EC2 instance and 200 GB of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region. Based on your choice, one Informatica high availability node with an EC2 instance and 200 GB of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region. Based on your choice, a bastion server with an EC2 instance that has 150 GB of storage space is created. The instance created is of type m5.large or m4.large, depending on availability of the instance size in the selected region.
IAM role and Instance Profile	1
Elastic IP	1 Elastic IP attached to the NAT gateway.

Components in a Deployment on an Existing VPC

The following table lists components that the cloud formation template creates when you deploy the software in an existing VPC:

Component	Number of Components Created
Security group	The following security groups: - One security group for the RDS - One security group for the Informatica domain instance - If you selected the option to create the bastion server, one security group to support remote login to the bastion server. - Based on your choice, two security groups for Informatica EMR workflow connections.
DB subnet group	1
RDS	1 Oracle RDS instance with 200 GB of storage capacity that runs on a db.m5.xlarge DB instance type.
EC2	Informatica domain with an EC2 instance and 200 GB of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region. Based on your choice, one Informatica high availability node with an EC2 instance and 200 GB of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region. Based on your choice, a bastion server with an EC2 instance that has 150 GB of storage space is created. The instance created is of type m5.large or m4.large, depending on availability of the instance size in the selected region.
AWS Lambda Function	5 - Lambda functions to check the feasability for VPC , Subnets and Informatica License
IAM role and Instance Profile	1

Bastion Server

You can optionally deploy a bastion server as an element in the solution. The bastion server acts as a firewall between the internet and the cloud platform network where the solution deploys. It can also act as a remote Windows server which you can log into to run Informatica clients.

Informatica domain

The Informatica domain is a server component that hosts application services, such as the Model Repository Service and the Data Integration Service. These services, together with domain clients, enable you to create and run mappings and other objects to extract, transform, and write data.

Application Services

Model Repository Service

The Model Repository Service manages the Model repository. The Model repository stores metadata created by Informatica products in a relational database to enable collaboration among the products. Informatica Developer, the Data Integration Service, and the Administrator tool store metadata in the Model repository.

Data Integration Service

The Data Integration Service is an application service in the Informatica domain that performs data integration tasks for the Developer tool and for external clients.

Metadata Access Service

The Metadata Access Service is an application service that allows the Developer tool to access Hadoop connection information to import and preview metadata. The Metadata Access Service contains information about the Service Principal Name (SPN) and keytab information if the Hadoop cluster uses Kerberos authentication.

Content Management Service

The Content Management Service is a component within the Informatica domain that manages reference data and is responsible for compiling rule specifications into mapplets. The Content Management Service provides reference data information to the Data Integration Service and to the Developer tool and Analyst tool. The Content Management Service stores reference data in a database that you specify.

Analyst Service

The Analyst Service is a component within the Informatica domain that runs the Analyst tool in the Informatica domain. The Analyst Service manages the connection between the service components and the users who log in to Analyst tool. You can perform column and rule profiling, manage scorecards, and manage bad records and duplicate records in the Analyst tool. The Analyst Service stores profiling, scorecarding, and bad and duplicate record data in databases that you specify.

Search Service

The Search Service is a component within the Informatica domain that manages search operations in the Analyst tool and Business Glossary Desktop. By default, the Search Service returns search results from a Model repository, such as data objects, mapping specifications, profiles, reference tables, rules, and scorecards.

Domain Repositories

Informatica repositories, posted on Oracle SQL databases, store metadata about domain objects. Informatica repositories include the following:

Domain configuration repository

The domain configuration repository stores configuration metadata about the Informatica domain. It also stores user privileges and permissions.

Model repository

The Model repository stores metadata for projects and folders and their contents, including all repository objects such as mappings and workflows.

In addition to these domain repositories, the solution also requires a repository for Hive metadata. This repository is hosted on an SQL database. It stores Hive table metadata to enable Hadoop operations.

For more information about domain repositories, see the Informatica Application Service Guide.

Informatica Clients

You can use several different clients with Data Engineering Streaming:

Administrator tool

The Administrator tool enables you to create and administer services, connections, and other domain objects.

Developer tool

The Developer tool enables you to create and run mappings and other objects that enable you to access, transform, and write data to targets.

Command line interface

The command line interface offers hundreds of commands to assist in administering the Informatica domain, creating and running repository objects, administering security features, and maintaining domain repositories.

Bastion server

You can optionally deploy a bastion server as an element in the solution. The bastion server is an AWS Windows instance installed with the Developer tool and the command line interface clients. The bastion server acts as a firewall for access to the VPC. It can also act as a remote Windows server that you can log into to run Informatica clients.

Before You Begin

Before you launch the automated deployment on the AWS marketplace, verify the prerequisites and make the choices described in this section.

Specialized Knowledge

Before you deploy Data Engineering Streaming, we recommend that you become familiar with the following AWS services.

If you are new to AWS, see Getting Started with AWS.

- Amazon VPC
- Amazon EC2
- Amazon RDS
- Amazon S3
- Elastic IP Addresses

If you are not familiar with Data Engineering Streaming and the Informatica platform, see the Informatica documentation portal.

Network Prerequisites

Before you deploy Data Engineering Streaming in the AWS cloud, verify that you have completed the following network prerequisites for the Informatica application services.

Network prerequisites depend on whether you choose to deploy the solution to a new subnet or an existing subnet.

When you deploy into a new VPC, the automated deployment builds a new AWS environment consisting of the VPC, subnets, NAT gateways, security groups, and other infrastructure components, and then deploys Data Engineering Streaming into this new VPC. When you deploy to an existing VPC, the automated deployment provisions Data Engineering Streaming in your existing AWS infrastructure.

New VPC Prerequisites

Verify the following requirements for all VPCs and subnets:

- Ensure that the selected availability zone has sufficient capacity to create a new subnet, route table, internet gateway, and NAT gateway.
- Ensure that the VPC creation has not exceeded the supported limit.

Existing VPC Prerequisites

In addition, observe the following prerequisites when you choose to deploy into an existing VPC:

- Ensure that the VPC has DNS resolution enabled.
- You must also attach the subnet to a route table with a local route to the VPC CIDR.
- Verify that each subnet in the VPC is associated with a route table.
- Verify that the subnet associated with an internet gateway has the Auto-assign Public IPv4 property enabled.
- Verify that each subnet, and each route listed in the route table, is defined as one of the following gateways:
  - Internet gateway
  - NAT gateway
  - Virtual private gateway
- For a successful deployment, verify that the service subnets you choose are attached to a route table under the selected VPC.
- Verify that your VPC has a minimum of two subnets. If you choose to create a bastion server, you can deploy it to one of the two subnets or use a separate public subnet.

Subnets require an Internet gateway, an NAT gateway, or a virtual private gateway in their route tables, to allow the instances to download packages and software without exposing them to the Internet. Record the domain name option configured in the DHCP options as explained in the Amazon VPC documentation. You provide the name in VPC settings when you configure the deployment.

Account and Security Prerequisites

Observe the prerequisites in this section to verify that you established the required AWS account and the requisite account private keys.

AWS Account and Region

If you don't already have an AWS account, create one at aws.amazon.com by following the on-screen instructions.

Use the region selector in the navigation bar to choose the AWS region where you want to deploy Data Engineering Streaming on AWS.

Note: The region you choose determines the availability of certain sizes of EC2 instances.

If necessary, request a service limit increase for the EC2 instance type you've decided to use for the Informatica domain. You might need to do this if you already have an existing deployment that uses this instance type, and you think you might exceed the default limit with this reference deployment.

Private Keys

Create a key pair in your preferred region.

When you log in to an EC2 instance, you use a password file for authentication. The file is called a private key file and has a file name extension of .pem.

If you do not have an existing .pem key to use, follow the instructions in the AWS documentation to create a key pair.

Note: Your administrator might ask you to use an existing key pair.

When you create a key pair, you save the .pem file to your desktop system. Simultaneously, AWS saves the key pair to your account. Make a note of the key pair that you want to use for the Data Engineering Streaming deployment, so that you can provide the key pair name during network configuration.

Account User Policies

Choose to grant the AWS user one of the following:

- Administrator privileges
- The following minimum policies attached to the user account:
  - arn:aws:iam::aws:policy/AWSCloudFormationFullAccess
  - arn:aws:iam::aws:policy/AmazonEC2FullAccess
  - arn:aws:iam::aws:policy/CloudWatchFullAccess
  - arn:aws:iam::aws:policy/IAMFullAccess
  - arn:aws:iam::aws:policy/AmazonS3FullAccess
  - arn:aws:iam::aws:policy/AmazonRDSFullAccess
  - arn:aws:iam::aws:policy/AmazonVPCFullAccess
  - arn:aws:iam::aws:policy/AWSLambdaFullAccess
  - arn:aws:iam::aws:policy/AmazonElasticMapReduceFullAccess
  - arn:aws:iam::aws:policy/AmazonElasticFileSystemFullAccess

Storage Prerequisite

The Data Engineering Streaming license must be stored in an Amazon S3 bucket.

Amazon S3 (Simple Storage Service) is an object repository in AWS. S3 stores objects, consisting of one or more files and accompanying file metadata, in "buckets." For information about S3, see the AWS documentation.

To use the automated cloud marketplace solution, you must create an S3 bucket to store the Data Engineering Streaming license. You can store the license on the bucket level or in a folder within the bucket.

Note the location of the license key and supply it when you configure the solution in the **License Key S3 Bucket URI** parameter.

Deploying Data Engineering Streaming on the AWS Marketplace

The automated deployment of Data Engineering Streaming on the AWS marketplace uses the AWS CloudFormation console to guide your choices and launch the solution deployment.

Use the AWS CloudFormation console to specify the stack name and to enter parameter values to the template that you use to deploy the application. A stack is a logical representation of an AWS deployment.

Specify the identifiers for the VPC, for the subnet that contains the Amazon EC2 instances where the deployment process deploys the application components, and for the subnet where the deployment process creates the Model repository. The deployment process provisions the application components and creates the Model repository based on the values that you specify.

Availability Zones and Subnets

The automated Amazon Marketplace solution deploys the components and services for Data Engineering Streaming into an Amazon VPC configured across two Availability Zones. Each Availability Zone is provisioned with two subnets.

In the first subnet, the deployment process provisions Data Engineering Streaming application services on an Amazon EC2 instance within the VPC:

- Data Integration Service
- Model Repository Service

- Metadata Access Service
- Content Management Service
- Analyst Service
- Search Service

For more information about domain components, see the Informatica Administrator Guide and Informatica Application Service Guide.

In the second subnet, the solution deploys an Oracle database on Amazon RDS for use as the Model repository. The Model repository stores metadata created by Informatica services.

Step 1. Get a License for Data Engineering Streaming

1. This deployment requires a license for Informatica Data Engineering Streaming. To sign up for a demo license, please contact Informatica.
2. Place the Data Engineering Streaming license key file for the software in the S3 bucket or directory that you created. You'll be prompted for its location in a later step.

Step 2. Launch the Deployment

You are responsible for the cost of the AWS services used while running this deployment. There is no additional cost for using this deployment. See the pricing pages for each AWS service you will be using for full details. Each deployment takes approximately one to two hours to complete.

Perform the following steps to launch the deployment:

1. Login to AWS Marketplace using the AWS account.
2. Type `Informatica Data Engineering Streaming` in the search box and press **Enter**.
3. Click **Informatica Data Engineering Streaming 10.4.0**. The **Product Overview** page appears.
4. Click **Continue to Subscribe**. The **Subscribe to this software** page appears.
5. Click **Continue to Configuration**. The **Configure this software** page appears.
6. Choose whether to deploy the solution on a new VPC or an existing VPC.
7. Click **Continue to Launch**. The **Launch this software** page appears.
8. Select **Launch CloudFormation** from the **Choose Action** drop-down list and click **Launch**. The **Create stack** page appears.
9. Verify the region for the deployment. The console displays the region in the upper-right corner of the navigation bar. Change the region if necessary.
10. In the **Stack** pane, ensure that the default option `Amazon S3 URL` is selected in the **Specify Template** section.

    Caution: Do not change any settings on this page or navigate away from the page. If you leave the page, the Amazon S3 URL might be erased.

    Click **Next**. The **Specify Stack Details** page opens.

Step 3. Specify Stack Details

On the **Specify stack details** page, review the parameters for the template. Provide values for the parameters that require input. For all other parameters, review the default settings and customize them as necessary.

1. Change the stack name if needed.
2. Supply values for the parameters in the **CloudFormation** screen.

You can use the information in the sections below to supply the values.

3. Under **Capabilities**, select the check box to acknowledge that the template will create IAM resources
4. To deploy the stack, click **Create**.

Amazon AWS begins to create the stack and displays the CloudFormation dashboard. When the status is CREATE_COMPLETE, the deployment is complete. You can use the URL displayed on the **Output tab** to view the stack properties.

Parameters to Deploy to a New VPC

When you choose to deploy the Data Engineering Streaming solution to a new VPC, you supply the following parameters on the **Specify Stack Details** page. Parameters marked with an asterisk (*) are mandatory.

Stack name

Enter a name for the stack. The stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).

Network Configuration

Enter values for the following parameters:

Parameter	Default	Description
Availability Zones*	Requires selections	List of Availability Zones to use for the subnets in the VPC. Multi-select in the dropdown list to choose at least two zones.
VPC CIDR*	Requires input	CIDR block for the new VPC. Format: `x.x.x.x/x` For example: `10.0.0.20/16`
Public Subnet CIDR*	Requires input	CIDR block for the public subnet. Must be within the VPC CIDR.
Private Subnet 1 CIDR*	Requires input	CIDR block for private subnet 1. Must be within the VPC CIDR.
Private Subnet 2 CIDR*	Requires input	CIDR block for private subnet 2. Must be within the VPC CIDR.
CIDR IP Range*	Requires input	The CIDR IP range that is permitted to access the Informatica domain. We recommend that you use a constrained CIDR range to reduce the potential of inbound attacks from unknown IP addresses. For example, to specify the range of 10.20.30.40 to 10.20.30.49, enter 10.20.30.40/49.

Bastion Server for Informatica

Enter values for the following parameters:

Parameter	Default	Description
Deploy Bastion Server*	No	Select Yes to deploy a bastion server to permit external clients to access resources in the VPC.

Amazon EC2 Configuration

Enter values for the following parameters:

Parameter	Default	Description
Key Pair Name*	Requires input	Select an EC2 key pair to enable SSH and RDP access to the Data Engineering Streaming solution.

Informatica Data Engineering Streaming Configuration

Enter values for the following parameters:

Parameter	Default	Description
License Key S3 Bucket URI*	Requires input	Enter the S3 Bucket URI for the Data Engineering Streaming license key. Use the following format: `s3://bucket-name/license.key` or `s3://bucket-name/sub-folder/license.key`
High Availability Deployment*	Disable	Choose to enable or disable high availability for the Data Engineering Streaming deployment. For information about high availability for the Informatica domain, see the Informatica documentation.
S3 Bucket Name	N/A	Name of an S3 bucket storage resource where source and target data reside. **Note:** This bucket is not the same as the bucket that contains the Data Engineering Streaming license key. Choose from the following options: - To use an existing bucket that contains source and target data, specify the name of a bucket in your account. - If you leave this entry field blank, no S3 bucket or connection are created. The automated deployment creates an S3 connection with the Informatica domain to enable you to query the data in the bucket. The IAM role for the EC2 instance is also associated with the bucket.
Cluster ID for Amazon EMR	N/A	ID for the Amazon EMR cluster used by the deployment. Choose to use an existing cluster or to create one. - To use an existing EMR cluster, enter the cluster ID. - To use the cluster that an Informatica workflow creates, leave the field empty. From 10.4.0, Intelligent Data Streaming supports cluster workflows. For more information, see Cluster Workflows.

Parameters to Deploy to an Existing VPC

When you choose to deploy the Data Engineering Streaming solution to an existing VPC, you supply the following parameters on the **Specify Stack Details** page. Parameters marked with an asterisk (*) are mandatory.

Stack name

Enter a name for the stack. The stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).

Network Configuration

Enter values for the following parameters:

Parameter	Default	Description
VPC*	Requires input	The identifier for the existing VPC
Subnet 1*	Requires input	The identifier for the subnet within the VPC where you want to deploy Data Engineering Streaming.
Subnet 2*	Requires input when High Availability is enabled.	The identifier for the subnet used by Data Engineering Streaming when High Availability is enabled. **Note:** This subnet must be in a different Availability Zone than Subnet 1.
CIDR IP Range*	Requires input	The CIDR IP range that is permitted to access the Informatica domain. We recommend that you use a constrained CIDR range to reduce the potential of inbound attacks from unknown IP addresses. For example, to specify the range of 10.20.30.20 to 10.20.30.24, enter 10.20.30.20/24.
Pre-validation Check*	No	Validates the VPC and subnets before deploying the Data Engineering Streaming solution. If you choose to have the automated solution perform a validation check before launching the deployment, the check validates your parameter inputs. If the deployment as specified fails the check, the deployment process stops. The deployment logs record the problem.

Bastion Server for Informatica

Enter values for the following parameters:

Parameter	Default	Description
Deploy Bastion Server*	No	Select Yes to deploy a bastion server to permit external clients to access resources in the VPC.
Bastion Subnet*	Requires input if you choose to deploy a bastion server	Select a subnet for the bastion server. The subnet must belong to the VPC that you specified for the bastion server. **Tip:** Choose a subnet that is attached to an internet gateway.

Amazon EC2 Configuration

Enter values for the following parameters:

Parameter	Default	Description
Key Pair Name*	Requires input	Select an EC2 key pair to enable SSH and RDP access to the Data Engineering Streaming solution.

Informatica Data Engineering Streaming Configuration

Enter values for the following parameters:

Parameter	Default	Description
License Key S3 Bucket URI*	Requires input	Enter the S3 Bucket URI for the Data Engineering Streaming license key. Use the following format: `s3://bucket-name/license.key` or `s3://bucket-name/sub-folder/license.key`
High Availability Deployment*	Disable	Choose to enable or disable high availability for the Data Engineering Streaming deployment. For information about high availability for the Informatica domain, see the Informatica documentation.
S3 Bucket Name	N/A	Name of an S3 bucket storage resource where source and target data reside. **Note:** This bucket is not the same as the bucket that contains the Data Engineering Streaming license key. Choose from the following options: - To use an existing bucket that contains source and target data, specify the name of a bucket in your account. - If you leave this entry field blank, no S3 bucket or connection are created, The automated deployment creates an S3 connection with the Informatica domain to enable you to query the data in the bucket. The IAM role for the EC2 instance is also associated with the bucket.
Cluster ID for Amazon EMR	N/A	ID for the Amazon EMR cluster used by the deployment. Choose to use an existing cluster or to create one. - To use an existing EMR cluster, enter the cluster ID. - To use the cluster that an Informatica workflow creates, leave the field empty. **Note:** Make the necessary changes for the inbound rules of the security group for EMR and the Informatica node. From 10.4.0, Intelligent Data Streaming supports cluster workflows. For more information, see Cluster Workflows.

Monitoring Instance Provision and Informatica Domain Creation

You can use cloud platform dashboards, logs, or other artifacts to see whether cluster creation succeeded and how to locate and identify the Informatica domain on the cloud platform.

Deployed AWS Resources

You can monitor the progress of the solution deployment in the Create Stack section of the AWS portal.

Each row of the display shows stack components and creation status. The stack is complete when the status reads CREATE_COMPLETE.

For deployment to a new VPC, the console displays the following stack components:

Component	Description
StackName-VPCStack	When you deploy to a new VPC, this template creates network-related resources, including the VPC, subnets, internet gateway, and the NAT gateway.
DESStack	The main stack.

In addition to the above components, the console displays the following stack components when you deploy to an existing VPC:

Component	Description
StackName/DESStack	Main stack. Always created.
StackName-LicenseCheckLambdaStack	Used to check if provided license key file is valid. Created by default.
tackName-BastionSubnetCheckLambdaStack	Used to check if a subnet is valid. Created by default. Lambda function will be created only when the following parameters are set to Yes: - **Pre-validation Check** - **Deploy Bastion Server**
StackName-SubnetsAZCheckLambdaStack	Used to check if a subnets are in different availability zones. Created by default.
StackName-PublicIpDnsCheckLambdaStack	Used to check if an auto-assign public IPv4 address is enabled for subnet. Created by default.
StackName-Subnet2ChecklambdaStack	Used to check if a subnet is valid. Default. Lambda function will be created if the **Pre-validation Check** parameter is set to Yes.
StackName-Subnet1CheckLambdaStack	Used to check if a subnet is valid. Default. Lambda function will be created if the **Pre-validation Check** parameter is set to Yes.
StackName-IAMStack	Creates IAM roles and profiles used by EC2 instances. Created by default.
StackName-SecurityGroupStack	Creates security groups and network interfaces used by EC2 instances. Created by default.
StackName-RDSStack	Creates RDS resources. Created by default.
StackName-BastionServerStack	Creates bastion server on an EC2 instance. Created only when you choose to create the bastion server
StackName-DESServerStack	Creates DES domain node on an EC2 instance. Created by default.
StackName-DESHANodeStack	Creates DES high availability node on EC2 instance. Created only when you choose to enable high availability

Output Tab Properties

Click the **Output** tab to view the list of resources that the deployment process created in the VPC. The table below lists the outputs:

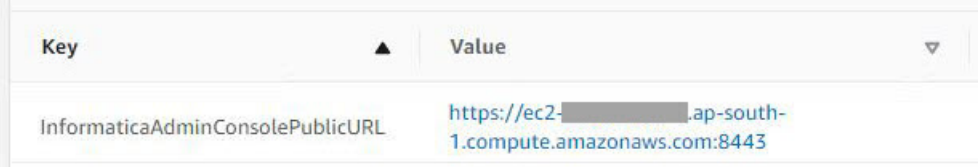
Output	Description
AdminConsoleUser	User ID of the Administrator tool client user.
AdminConsoleUserPwd	Password for the Administrator tool user.
BastionPrivateDNS	DNS of the private bastion server. For example: ip-<IPv4 address>-<region designation>.compute.internal
BastionPublicIPDNS	DNS of the public bastion server. For example: ip-<IPv4 address>-<region designation>.compute.amazonaws.com
DBUserPwd	Password for the database users of the domain databases created in the deployment.
DBUsers	User IDs for the database users of the domain databases created in the deployment.
HANodeIP	IPv4 address of the high availability or failover node for the domain.
InformaticaAdminConsolePrivateURL	URL:port of the Administrator tool hosted on a private server.
InformaticaAdminConsolePublicURL	Public URL:port of the Administrator tool.
InformaticaAdminConsoleServerLog	Path to the domain installation log.
InstanceSetupLogs	Path to the domain EC2 host setup log.

Use the URL, user ID, and password to access the Administrator tool to perform administrator tasks. For a description of domain administrator tasks, see the Informatica Administrator Guide and the Data Engineering Integration Administrator Guide.

Deployed Informatica Resources

The list of deployed resources includes the **InformaticaAdminConsolePublicURL**. Click the linked value of this property.

The following image shows an example of this property, with the IP address portion redacted:



The Administrator tool opens in a browser window.

Use the Informatica Administrator user name and password that are listed in the **Outputs** list to log in to the Administrator tool.

You can use the Administrator tool to view and administer the Informatica application services and resources that the automated deployment created.

Solution Deployment Logs

You can review the following logs to see the solution deployment as it happened. Review logs to see the success or failure of solution element creation.

Review the following logs in the `/var/log` directory:

- cfn-init.log
- cfn-init-cmd.log
- cloud-init.log

Review the following log in the `/opt` directory:

- Infa_OneClick_Solution.log

Informatica Domain Logs

You can access the following logs on the Informatica domain VM:

Command execution log

This log records the following events:

- Creation of Informatica connections, cluster configurations, and services.
- Population of the data warehouse and SQL databases.
- Data Integration Service recycling to register all changes.

At the top of the log file is a summary section that lists automated tasks and their status. Beneath the summary section are detailed sections about each task. If any of the tasks failed to complete successfully, you can look at the detailed section for the task to troubleshoot the task.

File name: `Infa_OneClick_Solution.log`

Location: `/opt`

CloudWatch logs for Informatica

CloudWatch is an AWS service that enables you to monitor all of the events in your AWS deployments. CloudWatch provides a central location from which to monitor EC2 instances and other resources. for more information, see the AWS documentation.

To monitor Data Engineering Streaming events in CloudWatch, go to the CloudWatch page and search for "Informatica."

If no log files are returned by the search, log into the VM where the Informatica domain is installed and check for logs with the filename "`cfn*.log`"

Access the Informatica deployment logs through the AWS console:

1. In the Amazon Web Services Console, select CloudWatch from the Services menu.
2. In the navigation pane, choose **Logs** > **Log groups**.
3. Enter `InformaticaDESLogGroup` in the search pane.

CloudWatch logs for Lambda functions

If the automated deployment fails during execution, CloudWatch logs display information about process errors.

Access the CloudWatch logs through the AWS console:

1. In the Amazon Web Services Console, select CloudWatch from the Services menu.
2. In the navigation pane, choose **Logs** > **Log groups**.

3. Enter `/aws/lambda/<stack name>` in the search pane.
4. Select the log group associated with the name of the stack that failed.

Authors

Harish R.G.

Saurabh Sugandh